Project Requirement Document (PRD)

Calculator & Currency Converter App

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1. Project Overview

Field	Details
Project Name	CalcMaster (Working Title)
Version	1.0
Platform	Android, iOS (Cross-platform)
Target Audience	Students, Professionals, General Users
Core Purpose	Provide a simple, intuitive calculator with integrated currency conversion, including support for Espees

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2. Objectives

- Deliver a **fast, clean, and easy-to-use** calculator for everyday and academic use.
- Provide a **reliable currency converter** with a broad range of global currencies and Espees.
- Ensure a **smooth onboarding experience** through walkthrough screens for first-time users.
- Maintain a **visually appealing dark grey and orange color theme** throughout the app.

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3. User Personas

Persona	Description
Student	Needs quick access to basic and scientific calculators for homework and exams.
Professional	Requires a streamlined tool for quick financial calculations and currency conversions.
General User	Everyday user looking for a clean, ad-free, reliable calculator app.

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4. Functional Requirements

4.1 Calculator Module

ID	Feature	Description
FR-01	Basic Calculator	Standard operations: addition, subtraction, multiplication, division, modulo.
FR-02	Scientific Calculator	Trigonometry, logarithms, exponents, percentages, square root, factorial, and more.
FR-03	History	Display recent calculations; allow users to tap and reuse previous results.
FR-04	Clear & Undo	Button to clear all input; ability to undo the last digit/operation entered.
FR-05	Keyboard Support	Optional keyboard input for power users.

4.2 Currency Converter Module

ID	Feature	Description
FR-06	Multi-Currency Conversion	Convert between major world currencies (USD, GBP, EUR, NGN, etc.).
FR-07	Espees Conversion	Convert Espees to and from other supported currencies.
FR-08	Live Exchange Rates	Fetch and display up-to-date exchange rates.
FR-09	Currency Selector	Dropdown or chip-based UI for selecting From and To currencies.
FR-10	Swap Currencies	Tap to reverse the conversion direction.

4.3 Onboarding Module

ID	Feature	Description
FR-11	Walkthrough Screens	3–4 sequential screens explaining: calculator use, currency conversion, and history access.
FR-12	Skip & Done Buttons	Users can skip walkthroughs or mark them as seen permanently.
FR-13	Auto-Skip on Return Visits	Walkthroughs appear only on the very first app launch.
FR-14	Accessible Help	In-app "How to Use" accessible from the settings menu for users who re-read it.

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5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	App should open in under 2 seconds. Calculator responses must be under 100ms.

NFR-02	UI/UX	Dark grey (#2D2D2D) and orange (#FF8C00) as the primary color theme. Clean, minimal design.
NFR-03	Accessibility	Support dynamic text sizes and screen reader compatibility.
NFR-04	Offline Support	Calculator works fully offline; currency converter works with cached rates.
NFR-05	Stability	App should not crash on any valid input; handle network failures gracefully.
NFR-06	Security	No user data stored without consent; secure API calls for exchange rates.

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6. Design Guidelines

6.1 Color Palette

Role	**Color**	**Hex Code**	**Usage**
Primary Background	Dark Grey	#2D2D2D	Main screen background
Secondary Background	Lighter Grey	#3A3A3A	Card/button backgrounds
Accent	Orange	#FF8C00	Buttons, highlights, icons
Text (Primary)	White	#FFFFFF	Main text
Text (Secondary)	Light Grey	#B3B3B3	Subtitles, descriptions

6.2 Walkthrough Design

- Full-screen swipeable cards.

- Large illustrations or icons.
- Short, easy-to-read descriptions.
- Orange accent for CTA buttons.
- Progress indicator (dots) at the bottom.

6.3 Iconography

- Simple, flat, consistent line-style icons.
- Orange accent for active/highlighted icons.
- White/light grey for default icons.

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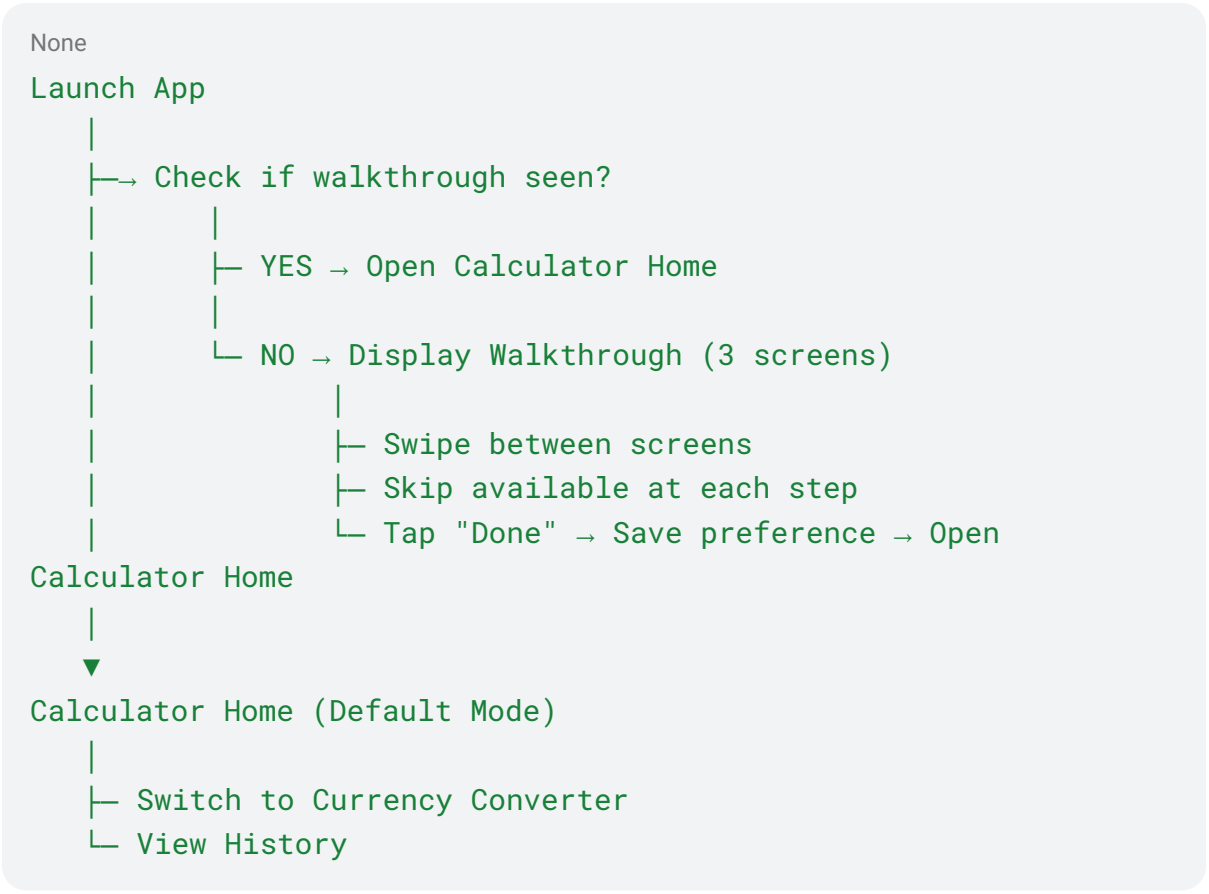
7. Technical Requirements

Area	**Requirement**
Framework	Flutter (for cross-platform deployment) or React Native.
Backend	REST API for live currency exchange rates (e.g., Exchangerate API, Frankfurter).
Espees API	Integration with an Espees-specific rate provider or manual rate management.
Local Storage	SQLite/SharedPreferences for history and onboarding preference.
State Management	Provider, Riverpod, or equivalent.
Testing	Unit tests for calculator logic; UI tests for walkthroughs and conversion flow.

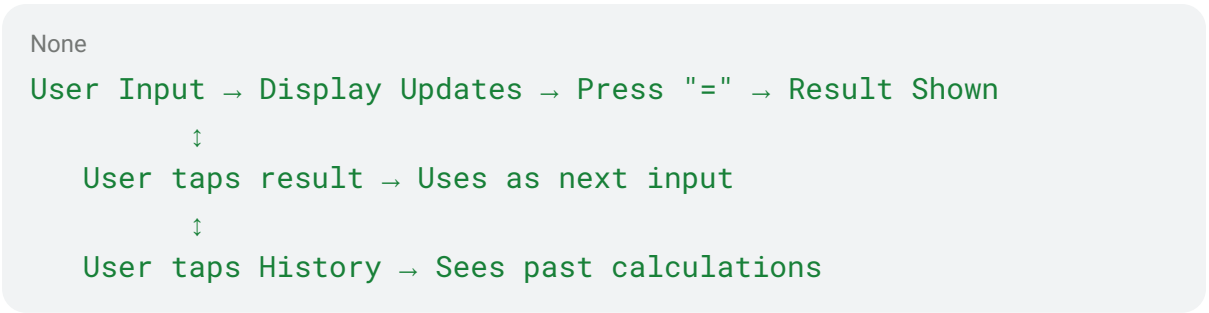
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8. User Flow

8.1 First-Time User Flow



8.2 Calculator Flow



8.3 Currency Converter Flow

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None
Open Converter → Select From Currency → Select To Currency →
Enter Amount → Display Result
        |
        └→ Tap Swap → Reverse directions
        └→ Tap History → See previous conversions
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9. Milestones

Phase	Description	Timeline
Phase 1	UI/UX Design & Prototyping	Week 1–2
Phase 2	Calculator Module Development	Week 3–5
Phase 3	Currency Converter Module + Espees Integration	Week 6–8
Phase 4	Walkthrough Onboarding & Settings	Week 9
Phase 5	Testing, Bug Fixes, and Polish	Week 10
Phase 6	Deployment to App Stores	Week 11

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10. Risks & Mitigations

Risk	Impact	Mitigation
Exchange rate API downtime	High	Cache last known rates; fallback to offline mode.
Espees data unavailable	Medium	Provide manual rate update option; partner with Espees provider.
Low user retention (no re-visits)	Low	Push notifications for rate alerts (optional, user opt-in).

Complex scientific calculator	Low	Progressive disclosure; hide advanced functions behind toggle.

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11. Future Enhancements (Roadmap)

- **Widget support** – Calculator widget for home screen.
- **Dark/Light mode toggle** – Let users switch themes.
- **Unit Converter** – Length, weight, temperature, etc.
- **Bookmark currencies** – Save frequently used pairs.
- **Notifications** – Alert when currency rates hit a target threshold.

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12. Success Metrics

Metric	**Target**
Average session duration	> 2 minutes
Walkthrough completion rate	> 70%
App Store Rating	> 4.5 stars
Crash-free sessions	> 99%
Active users (Month 1)	> 1,000

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I hope this PRD gives you a strong, clear blueprint to kick off your app project! If you'd like me to tailor it further—say, adding wireframe descriptions, refining the Espees integration

details, or converting this into a presentation-ready format—just say the word. You're building something useful, and I'm here to help you make it excellent.